

Abstract No.

Local strategy to remodel the glioblastoma microenvironment and “open the door” to anti-tumoral immune cells

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Glioblastoma (GB) is a highly aggressive brain tumor associated with extremely poor prognosis. The failure of current curative strategies is driven by a marked hypoxic and immunosuppressive tumor microenvironment along with recurrence. Our objective is to develop a novel therapeutic approach based on a hyaluronic acid (HA) hydrogel engrafted with a selective chemokine designed to stimulate anti-tumor immune responses and contribute to cognitive recovery following tumor resection. First, through gene signatures analyses from TCGA bulk RNA-seq database, we identified absence of markers of early, immature and mature tertiary lymphoid organs (TLOs) in GB including some chemokines involved in the establishment of mature TLO structures. We here investigated the therapeutic potential of a selective chemokine delivered within the cavity resection through our thermosensitive HA-hydrogel.

Following murine GB resection in immunocompetent mice, the HA-based hydrogel containing a free chemokine significantly reduced tumor recurrence volume, promoted MECA79+ high endothelial venule (HEV)-like immature structures and increased immune cell infiltration notably CD20+ B-cell recruitment. while improving mouse survival. This response was associated with the formation of B-cell aggregates in the pericavity tissue and infiltration of B cells into the dura mater characterized thanks to an original whole-head imaging protocol combined with a 3D bone-clearing approach preserving the skull bone. This currently allows evaluation of our strategy on vascular architecture remodeling and TLOs formation.

Finally, to test the resection cavity refunctionalization, we led longitudinal behavioral analyses comparing GB, resection+HA and resection+HA+chemokine mice and showed that chemokine restored anxiety-related behavior and improved cognitive performance, both altered by the tumor. Together, these findings demonstrate that local delivery of a key chemokine through a thermosensitive HA-hydrogel is a promising therapeutic strategy against glioblastoma. Supported by Ligue contre le Cancer Normandie, Région Normandie et les Fonds Européens du Développement Régional (FEDER), Géfluc, Université de Rouen Normandie and Inserm.



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